Final Project: Conduct the NIST Rhino Hunt

Completion requirements

To do: Make a submission

Due: Sunday, 8 September 2024, 11:00 PM

Conduct the NIST Rhino Hunt

Introduction

The National Institute of Standards and Technology provides a set of practice forensic images that can be freely downloaded and used to hone your forensic skills. You can find the full set at www.cfreds.nist.gov.

In this project we will use

- the Rhino hunt scenario and
- the SANS SIFT image

Project Steps:

Step 1: Download SIFT Workstation

Step 2: Import SIFT Workstation to VMware and open it.

Step 3: Login to SIFT Workstation

Step 4: Download the SANS Rhino hunt (File DFRWS2005-RODEO.zip is available in the download section below)

Step 5: Unzip the Rhino hunt

Step 6: Use SIFT to find the rhino pictures.

Step 7: Create a directory for the output:

Step 8: Run foremost against the RHINOUSB image.

Step 9: Review the output.

To open the file you have recovered, click the filing cabinet icon at the top left of the screen, navigate to Home ➢ Output ➢ Doc, and then double-click on the DOC file you recovered. Read to the end of the file.

Deliverable

1. Take a screenshot of each step in this Project and submit in one pdf document.
2. Answer the question: What happened to the hard drive? Once you know where the hard drive went, you are done with this project.

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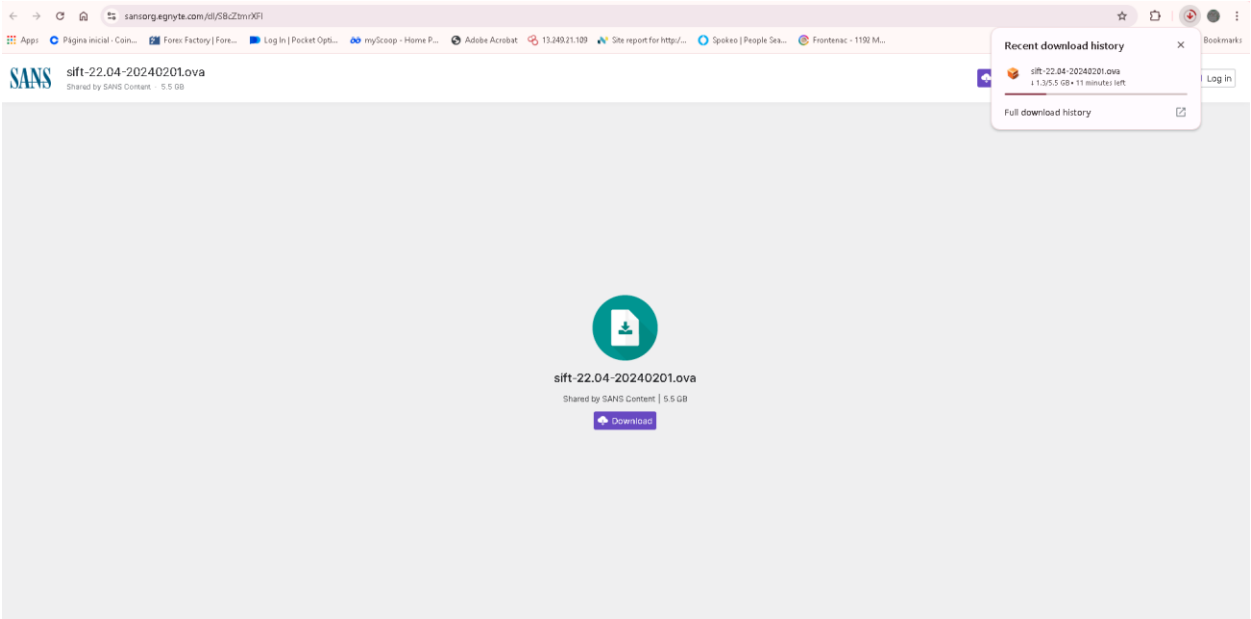
Please read before uploading the project:

- Submit file should be in PDF or World DOC only. Any other file format will not be accepted.
- Project to be submitted as ONE file only.
- Project should show full details: What is required, detailed steps of what you did.
- Screenshots for above steps.

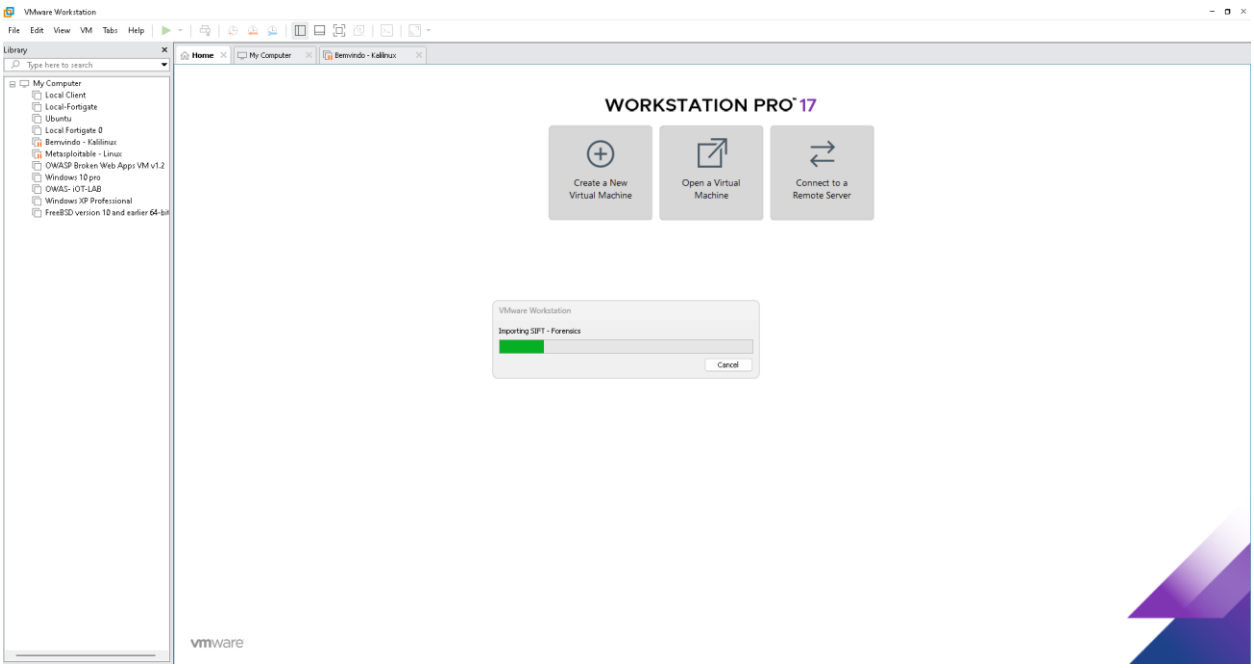
The Rhino hunt has a lot more to it, so feel free to continue based on the NIST page's instructions*

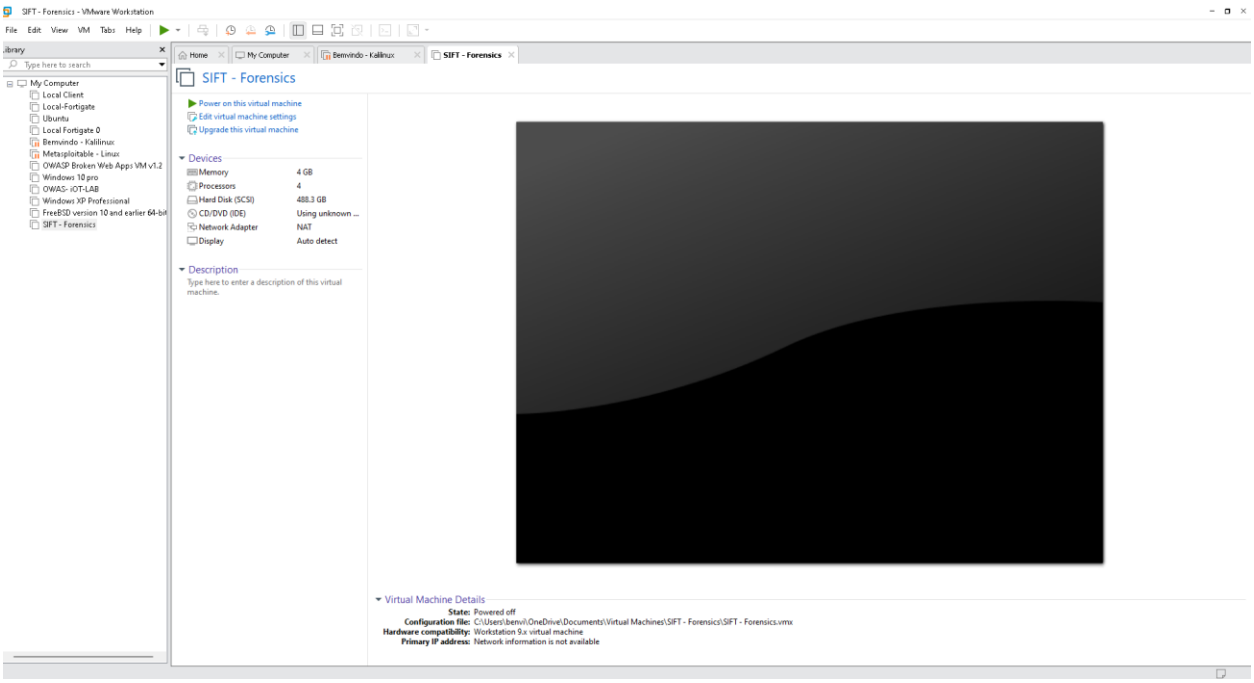
I went to the SANS website and downloaded the SIFT Workstation.

SIFT Workstation is a pre-configured Ubuntu-based virtual machine that includes a variety of forensic tools.

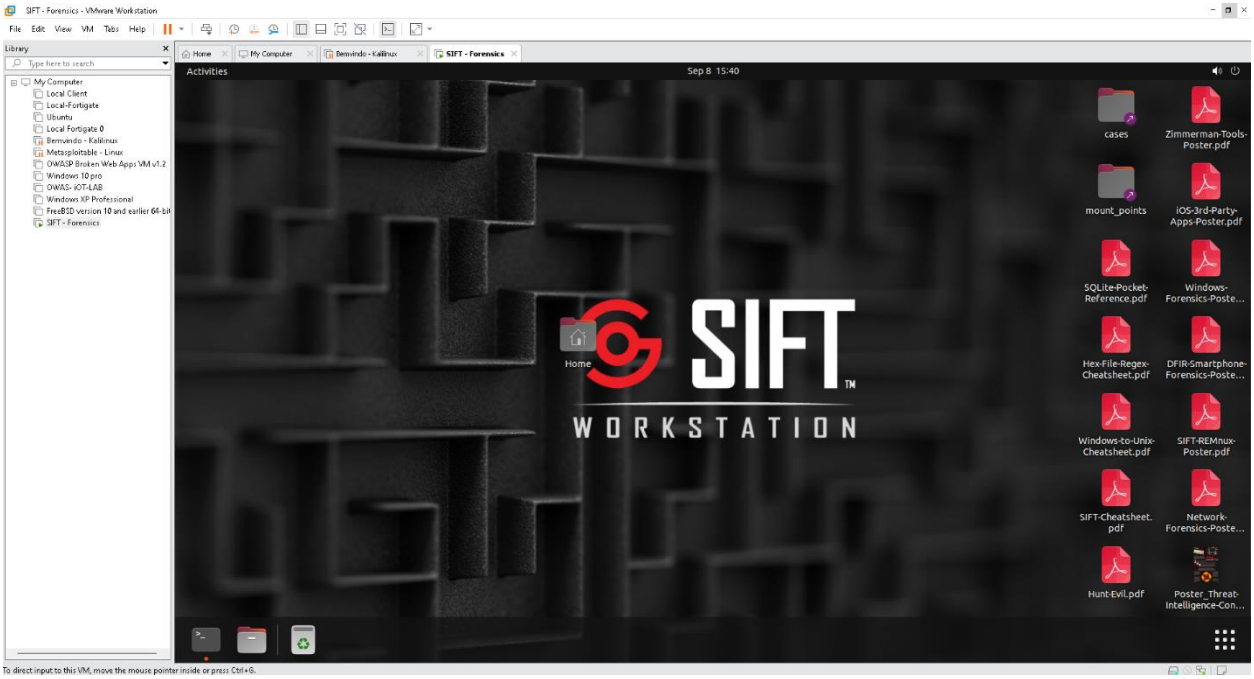


- Once downloaded, open VMware and selected ISO Download and I Opened, and locate the SIFT Workstation ova file.
- Start the virtual machine, allowing it to boot up.



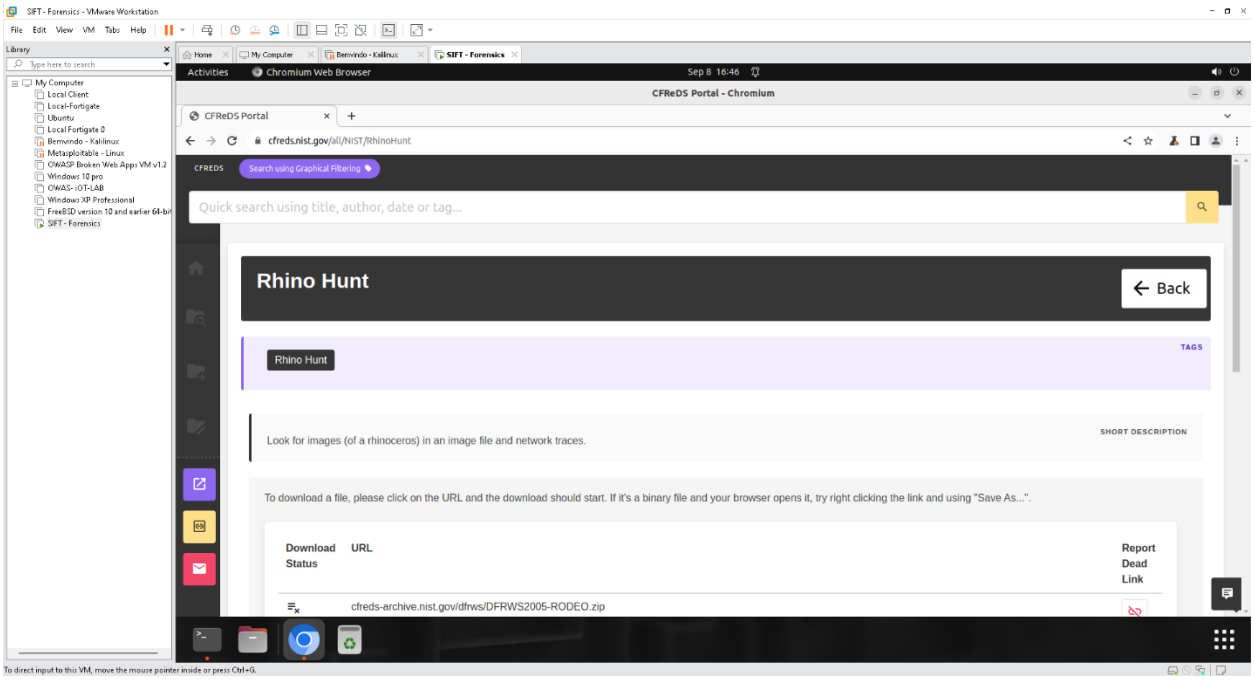


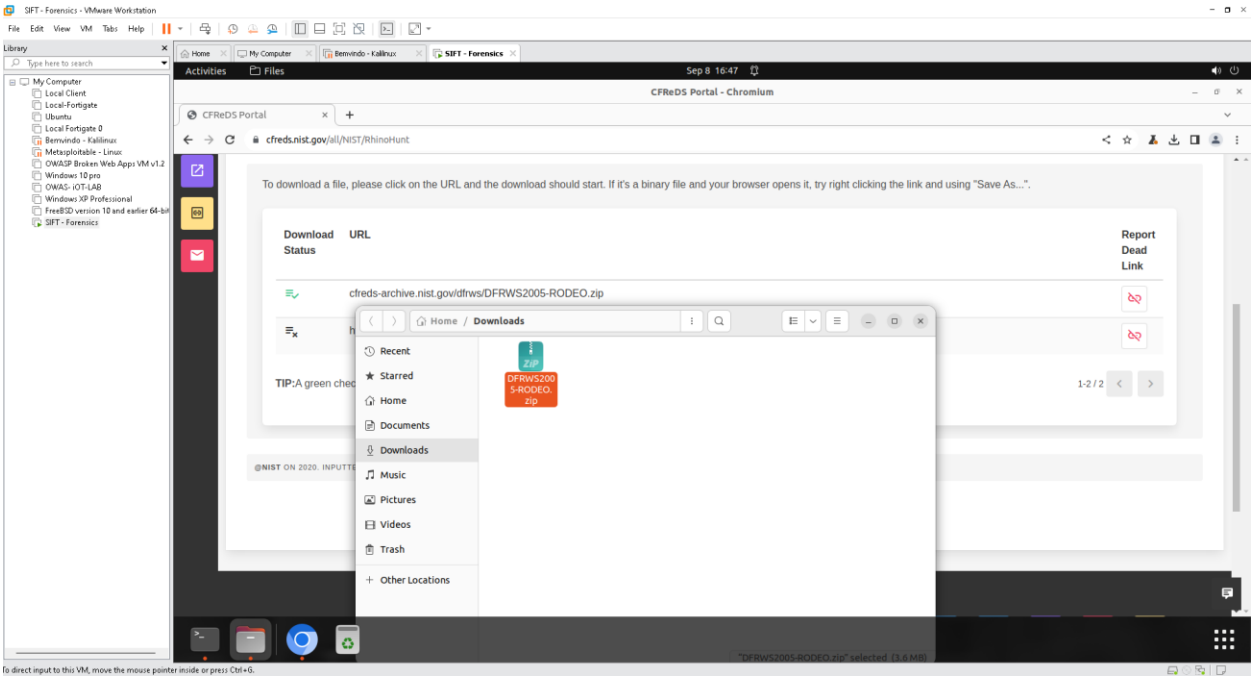
I entered the Default login credentials which is usually username: **sansforensics,** password: **forensics** and later I changed the credentials for security reasons



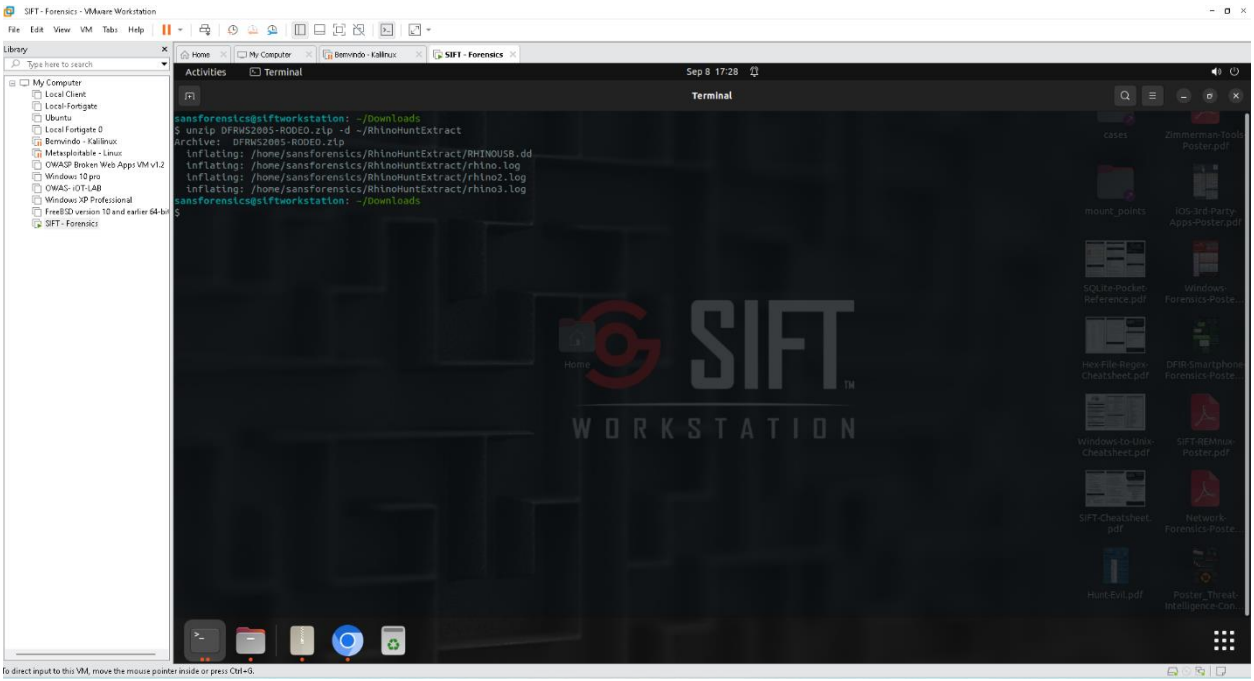
I went to the CFReDS website and find the Rhino Hunt scenario.

Download the DFRWS2005-RODEO.zip file into my SIFT virtual machine.





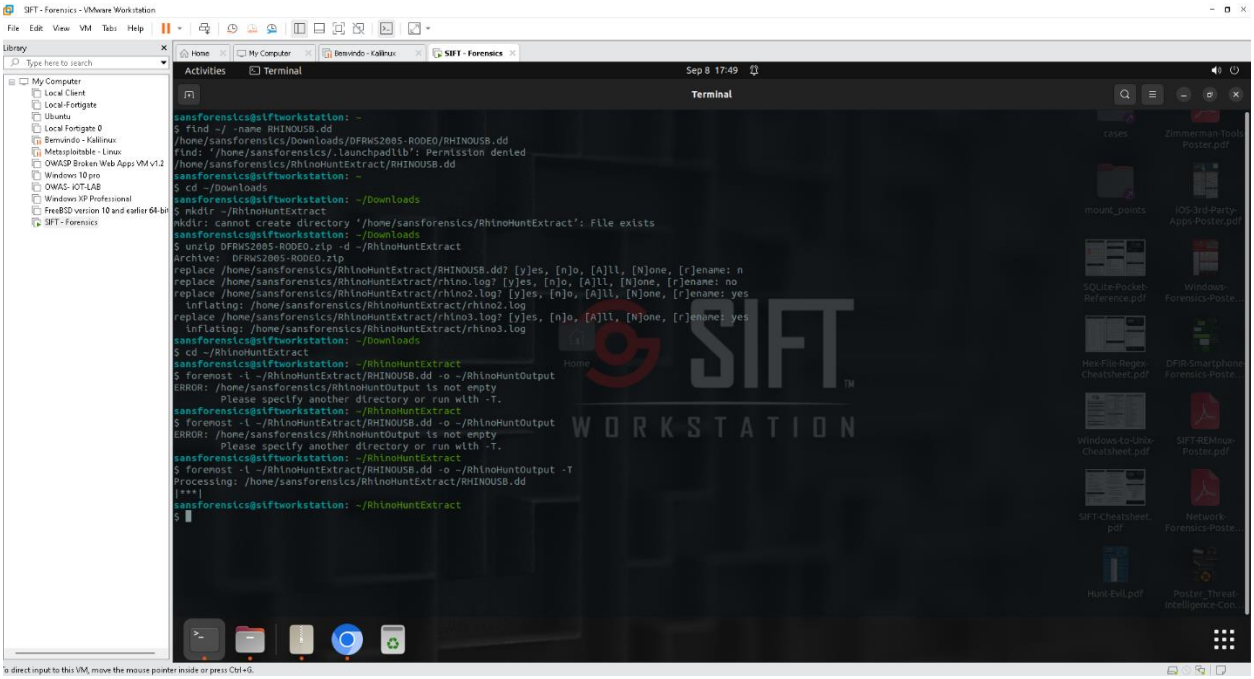
Used the unzip command to extract the contents of the DFRWS2005-RODEO.zip file:

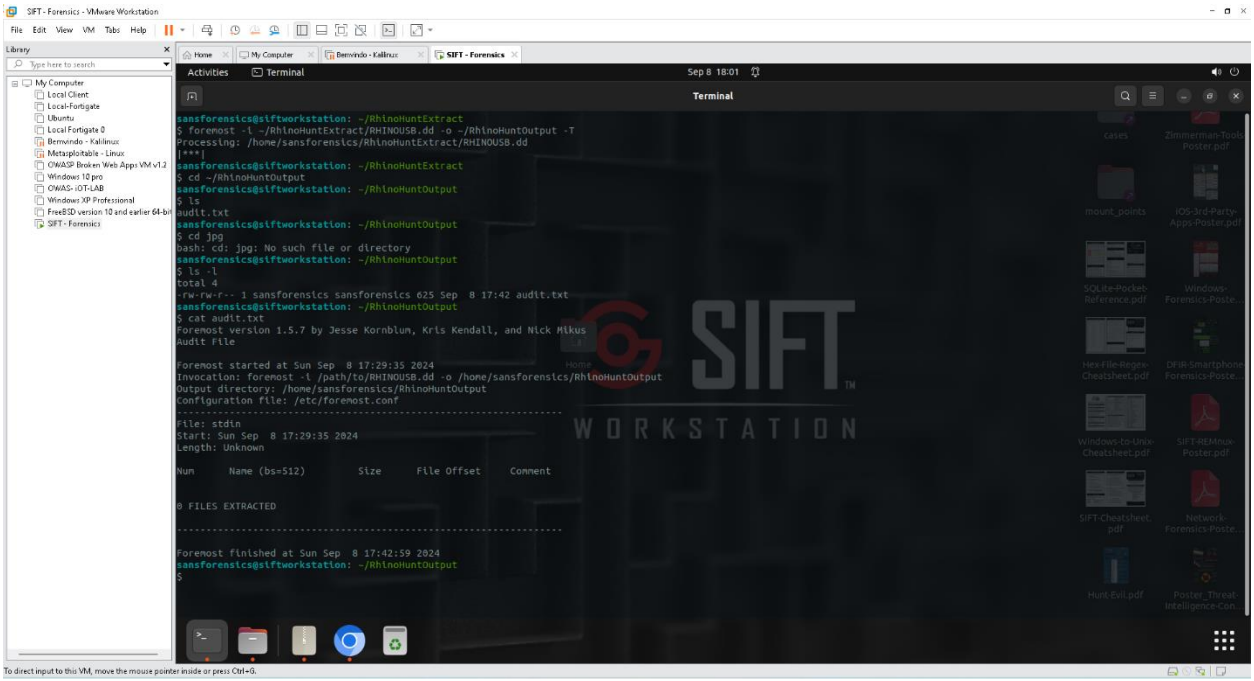


I analyzed the Rhino USB image which should be included in the extracted files

Start by creating a directory to hold the output

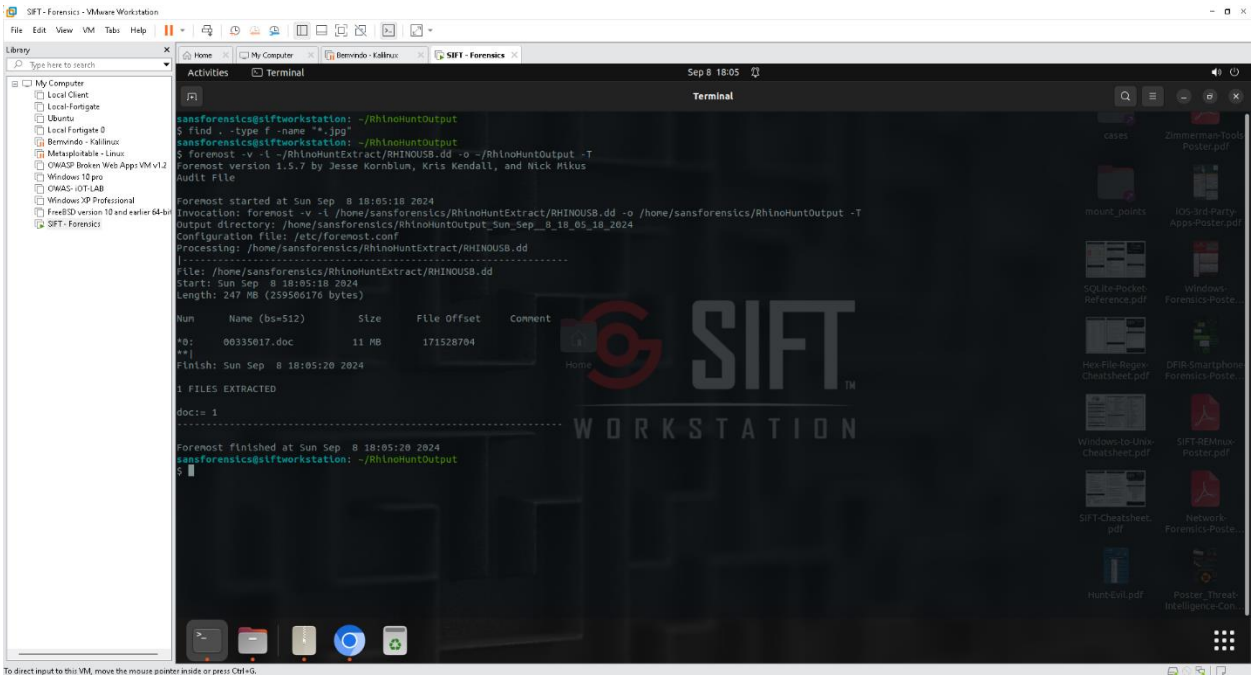
In the terminal, created a directory for storing extracted files from the Rhino USB image



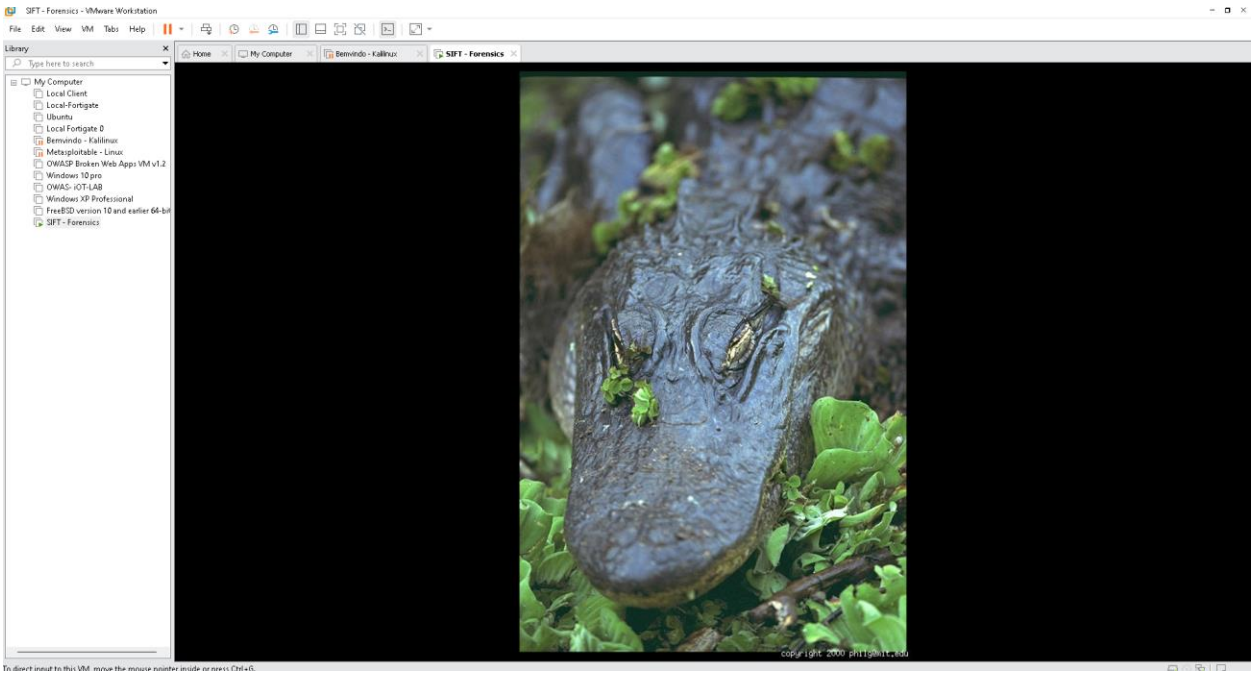

SIFT - Forensics - VMware Workstation
Activities
Terminal
Sep 8 18:01
sansforensics@siftworkstation: ~/RhinoHuntExtract
$ foremost -i ~/RhinoHuntExtract/RHINOUSB.dd -o ~/RhinoHuntOutput -T
Processing: /home/sansforensics/RhinoHuntExtract/RHINOUSB.dd
|***|
sansforensics@siftworkstation: ~/RhinoHuntExtract
$ cd ~/RhinoHuntOutput
sansforensics@siftworkstation: ~/RhinoHuntOutput
$ ls
audit.txt
sansforensics@siftworkstation: ~/RhinoHuntOutput
$ cd jpg
bash: cd: jpg: No such file or directory
sansforensics@siftworkstation: ~/RhinoHuntOutput
$ ls -l
total 4
-rw-rw-r-- 1 sansforensics sansforensics 625 Sep 8 17:42 audit.txt
sansforensics@siftworkstation: ~/RhinoHuntOutput
$ cat audit.txt
Foremost version 1.5.7 by Jesse Kornblum, Kris Kendall, and Nick Mikus
Audit File
Foremost started at Sun Sep 8 17:29:35 2024
Invocation: foremost -i /path/to/RHINOUSB.dd -o /home/sansforensics/RhinoHuntOutput
Output directory: /home/sansforensics/RhinoHuntOutput
Configuration file: /etc/foremost.conf

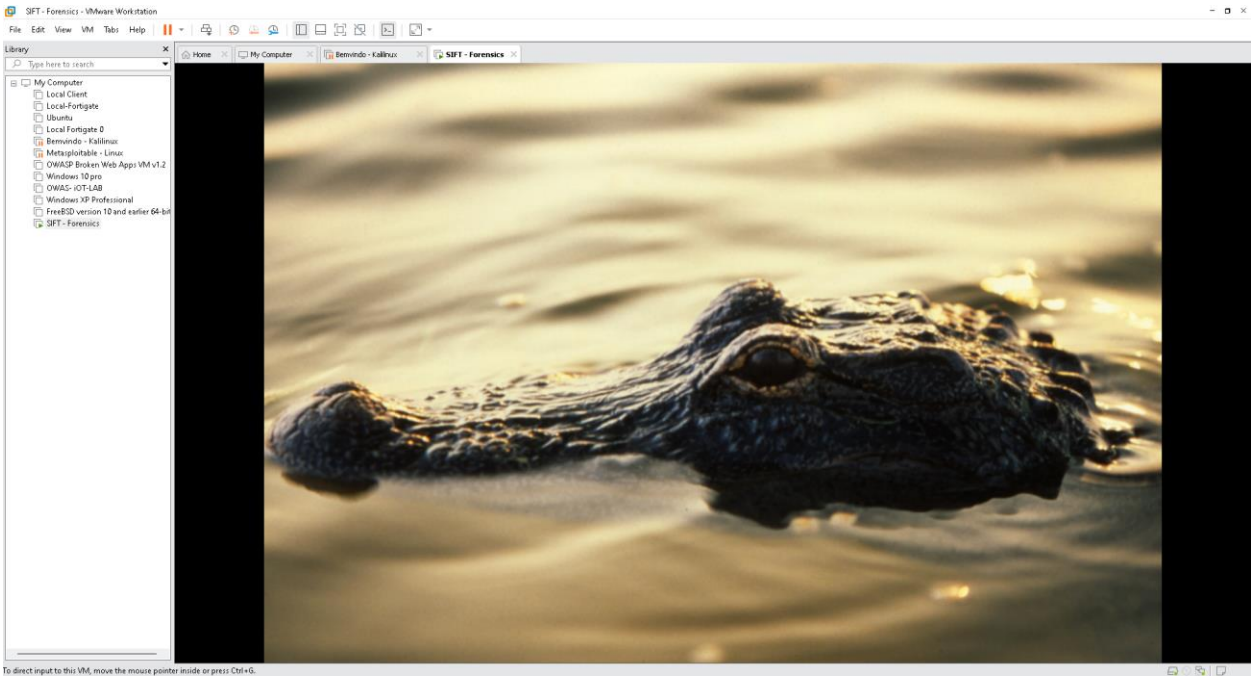
File: stdin
Start: Sun Sep 8 17:29:35 2024
Length: Unknown
Num Name (bs=512) Size File Offset Comment
0 FILES EXTRACTED

Foremost finished at Sun Sep 8 17:42:59 2024
sansforensics@siftworkstation: ~/RhinoHuntOutput
$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


SIFT - Forensics - VMware Workstation
Activities
Terminal
Sep 8 18:05
sansforensics@siftworkstation: ~/RhinoHuntOutput
$ find . -type f -name "*.jpg"
sansforensics@siftworkstation: ~/RhinoHuntOutput
$ foremost -v -i ~/RhinoHuntExtract/RHINOUSB.dd -o ~/RhinoHuntOutput -T
Foremost version 1.5.7 by Jesse Kornblum, Kris Kendall, and Nick Mikus
Audit File
Foremost started at Sun Sep 8 18:05:18 2024
Invocation: foremost -v -i /home/sansforensics/RhinoHuntExtract/RHINOUSB.dd -o /home/sansforensics/RhinoHuntOutput -T
Output directory: /home/sansforensics/RhinoHuntOutput_Sun_Sep__8_18_05_18_2024
Configuration file: /etc/foremost.conf
Processing: /home/sansforensics/RhinoHuntExtract/RHINOUSB.dd
|
File: /home/sansforensics/RhinoHuntExtract/RHINOUSB.dd
Start: Sun Sep 8 18:05:18 2024
Length: 247 MB (259506176 bytes)
Num Name (bs=512) Size File Offset Comment
*0: 00335017.doc 11 MB 171528704
**|
Finish: Sun Sep 8 18:05:20 2024
1 FILES EXTRACTED
doc:= 1

Foremost finished at Sun Sep 8 18:05:20 2024
sansforensics@siftworkstation: ~/RhinoHuntOutput
$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


SIFT - Forensics - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Local Client
Local-Fortigate
Ubuntu
Local Fortigate 0
Bemvindo - Kalilinux
Metasploitable - Linux
OWASP Broken Web Apps VM v1.2
Windows 10 pro
OWAS- IOT-LAB
Windows XP Professional
FreeBSD version 10 and earlier 64-bit
SIFT - Forensics
Home
My Computer
Bemvindo - Kalilinux
SIFT - Forensics
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


SIFT - Forensics - VMware Workstation
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My Computer
Local Client
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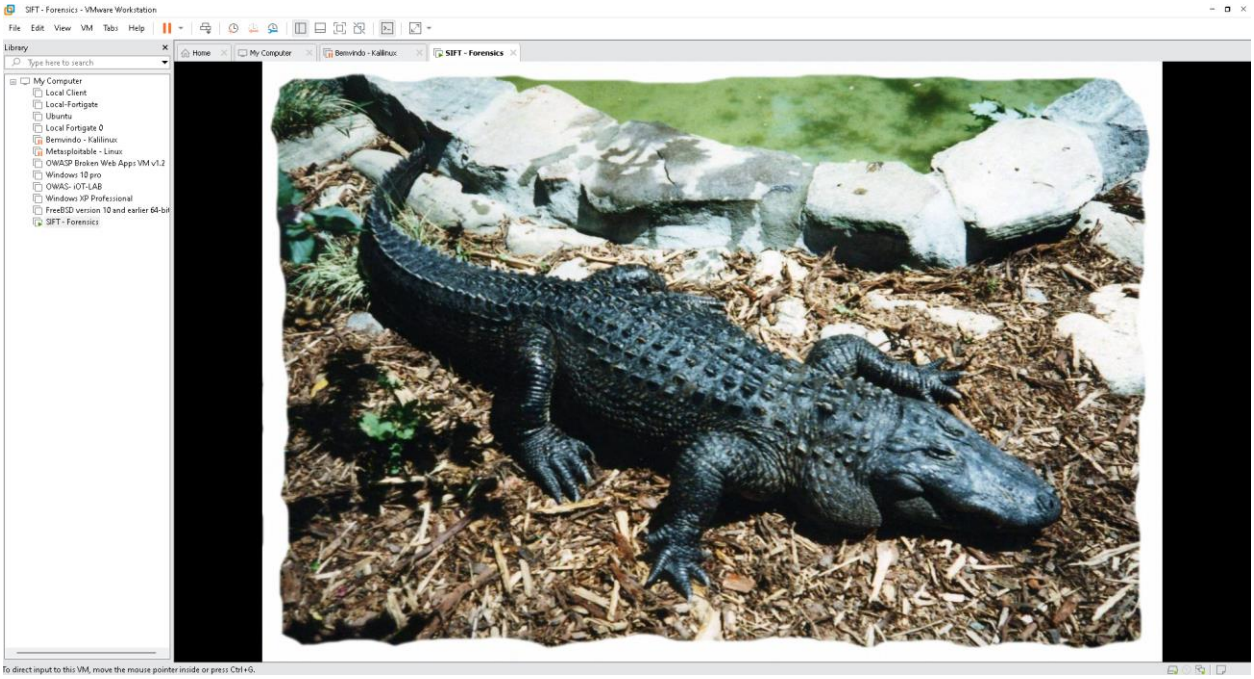
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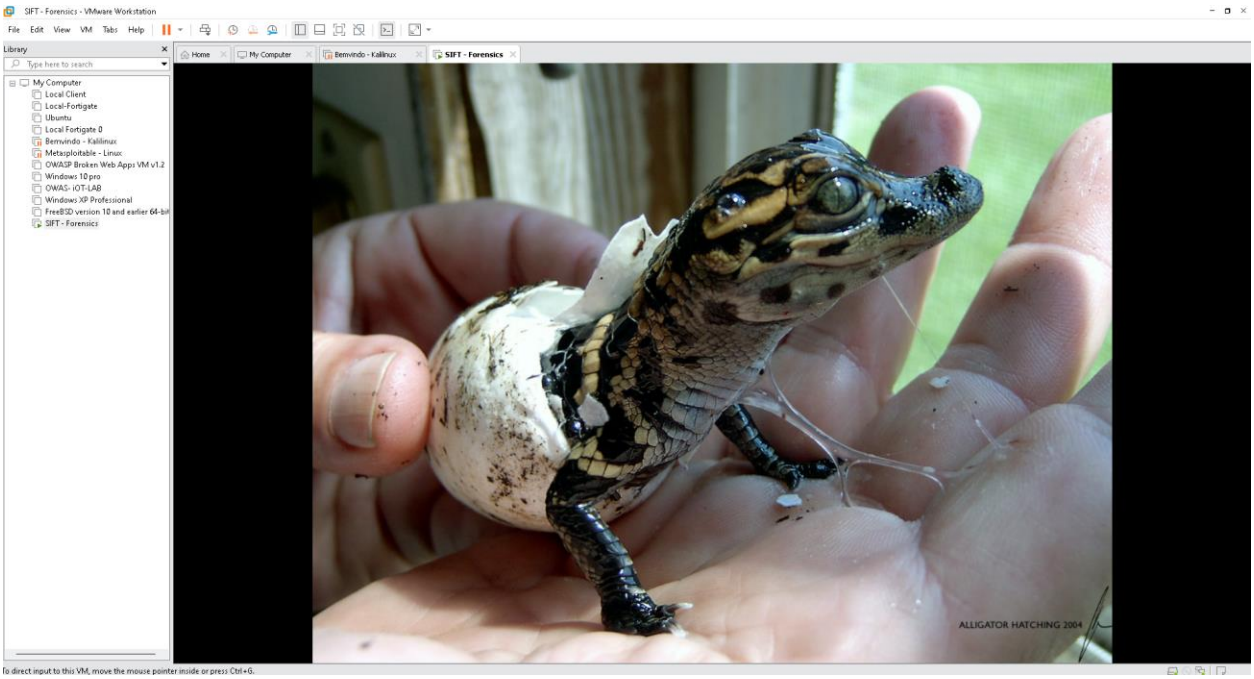
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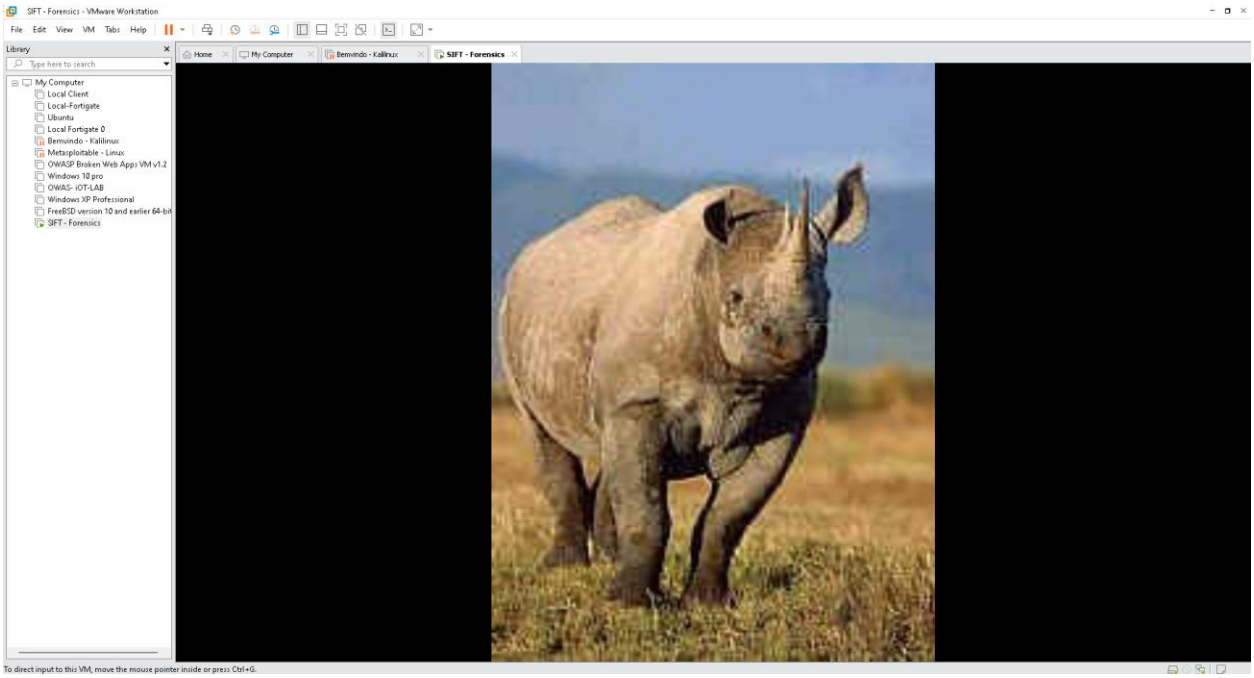
SIFT - Forensics
ALLIGATOR HATCHING 2004
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.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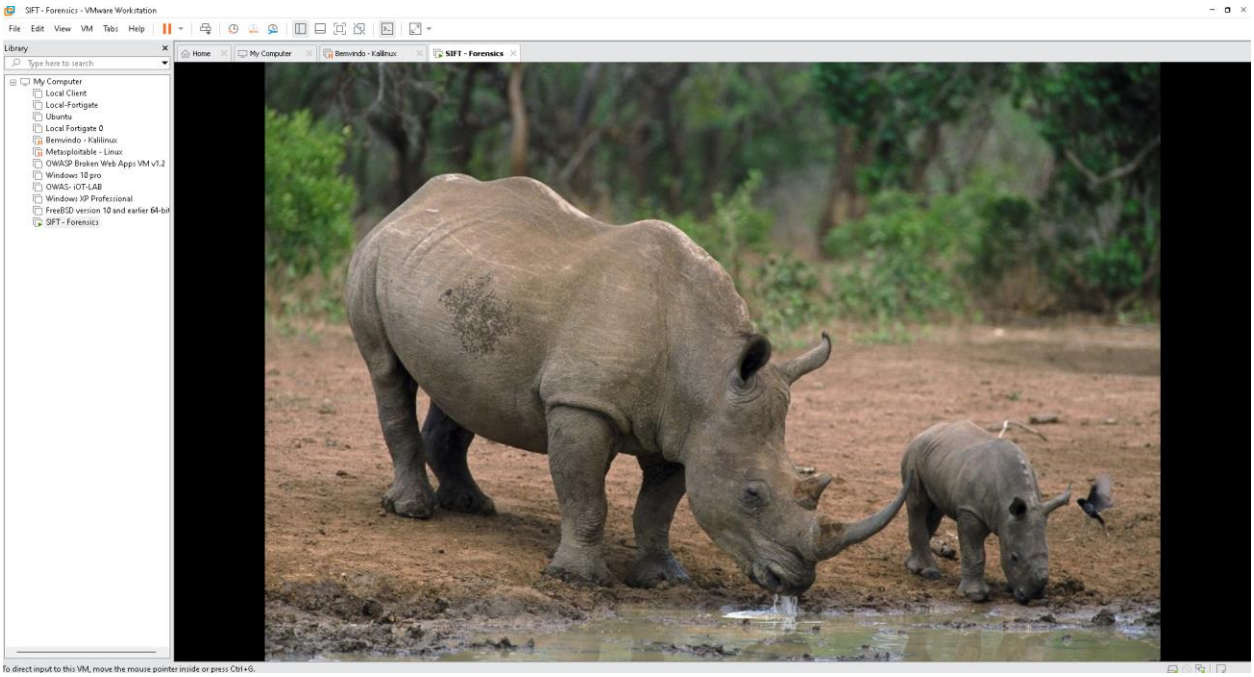
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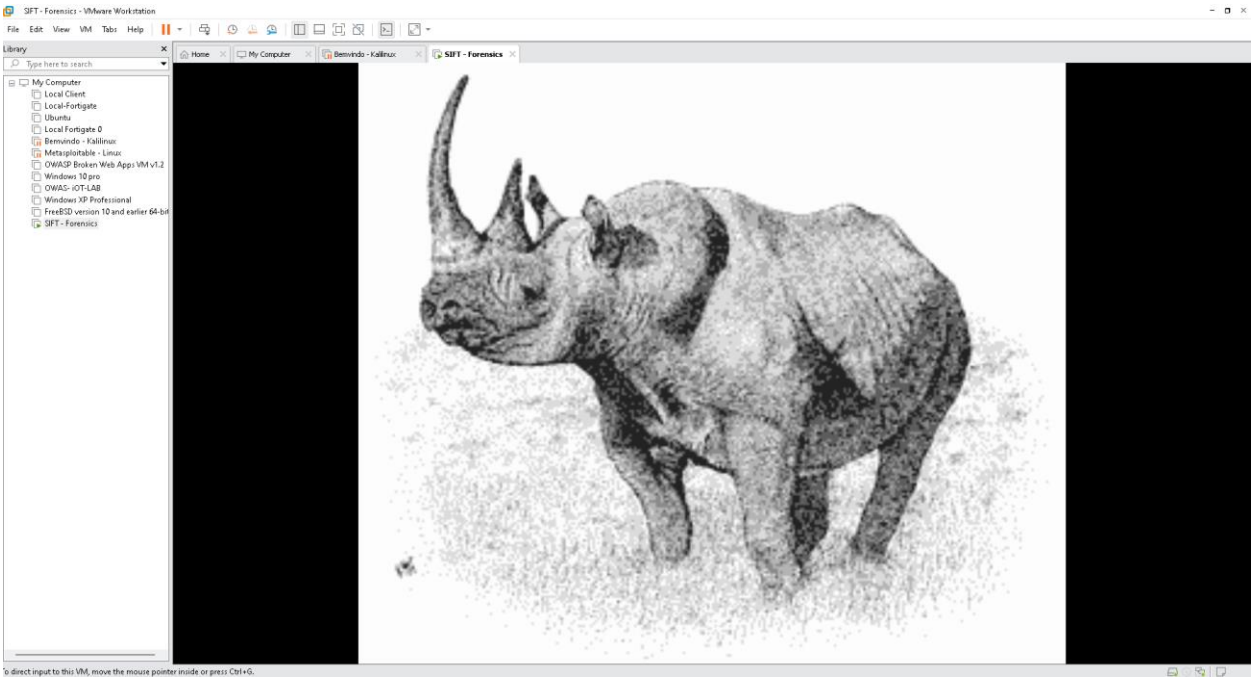
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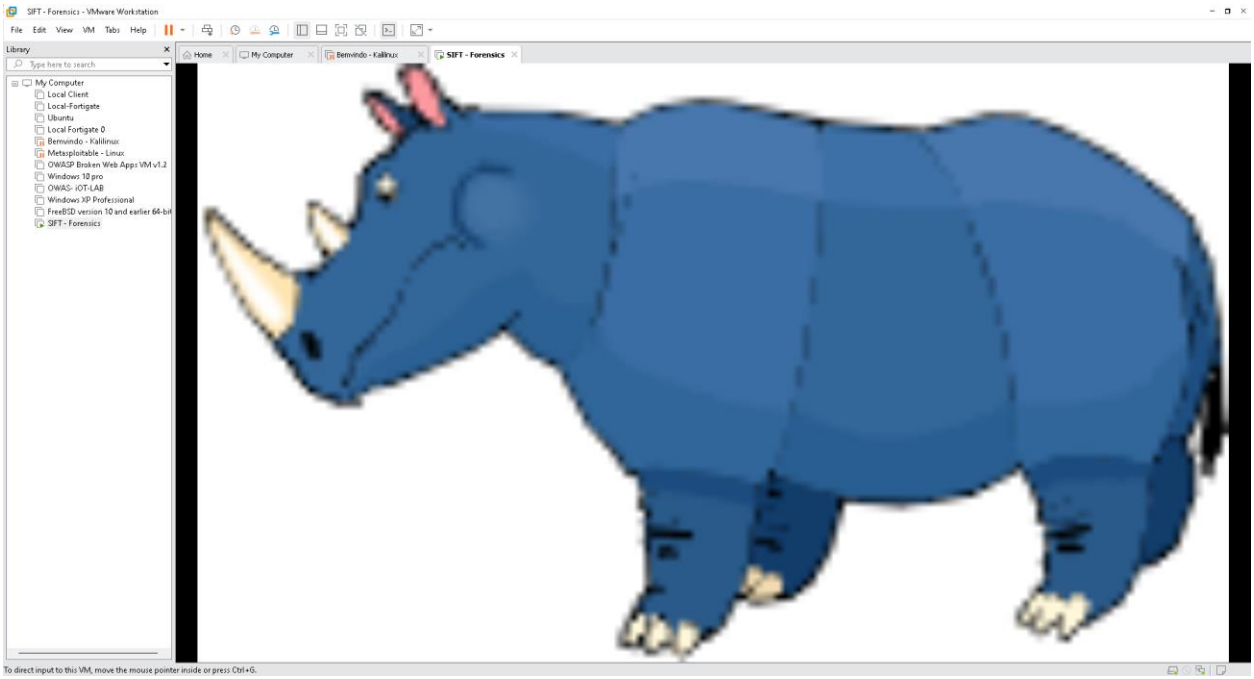
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She died in February at the age of 74. In August 2001 it wasn't a decision, since the alternative was regret. It wasn't her fault that I didn't go to the drugstore... And then getting her to arrange a time with Lynn, so that I can tell her just with me and Tal there.

We were walking from the restaurant to the Irish pub, and who did we run into? Then we had dinner at this really nice restaurant with a patio kind of in Old Town.

Back in March I did a presentation at a research conference held at UC Irvine and presented by the Honors Transfer Council of California.

My mother and I have a unique relationship. Chasing Amy - *whimpers* By all accounts i should have liked this film. Their relationship was a failure! All other IB families are trying to keep their kid in IB.. trying to encourage their kid to do good.. mine is trying to make me quit. Anyway, this one is someone she was involved with in high school who says he's been trying to find her all these years and finally tracked her down. So seeing how I am scared of pitch black darkness I got up and was trying to see what made the power go out, and my parents got up and joined me with flashlights and candles. I handed in the damn homework, which was really quite stupid for all those who do know how to use the damn computer. He turned back to me, completely sober, completely serious and replied - 'No, I don't dislike you, I'm just scared of you.' I stared at him in disbelief for a moment or two or three - scared of me?

There truly are buckets of phenomenal things to be amazed by, and thankful for.

Stocklos and Andrew were only there for one of the nights I was.

Then there was one 4th of July when we had just moved to a big new house in the country & we were flat busted, so we had no funds to celebrate. But Jon and been watching the encounter and remained at bay purposely in order to let me say what I had to say to Andrew. I am so excited about the trip... and so excited that Jen and Nicole are going... and so excited that we are going to both Wolves games while we are there. And she attempted lying to me about going Hong Kong together. And look, it's very difficult to work both sides of the aisle.

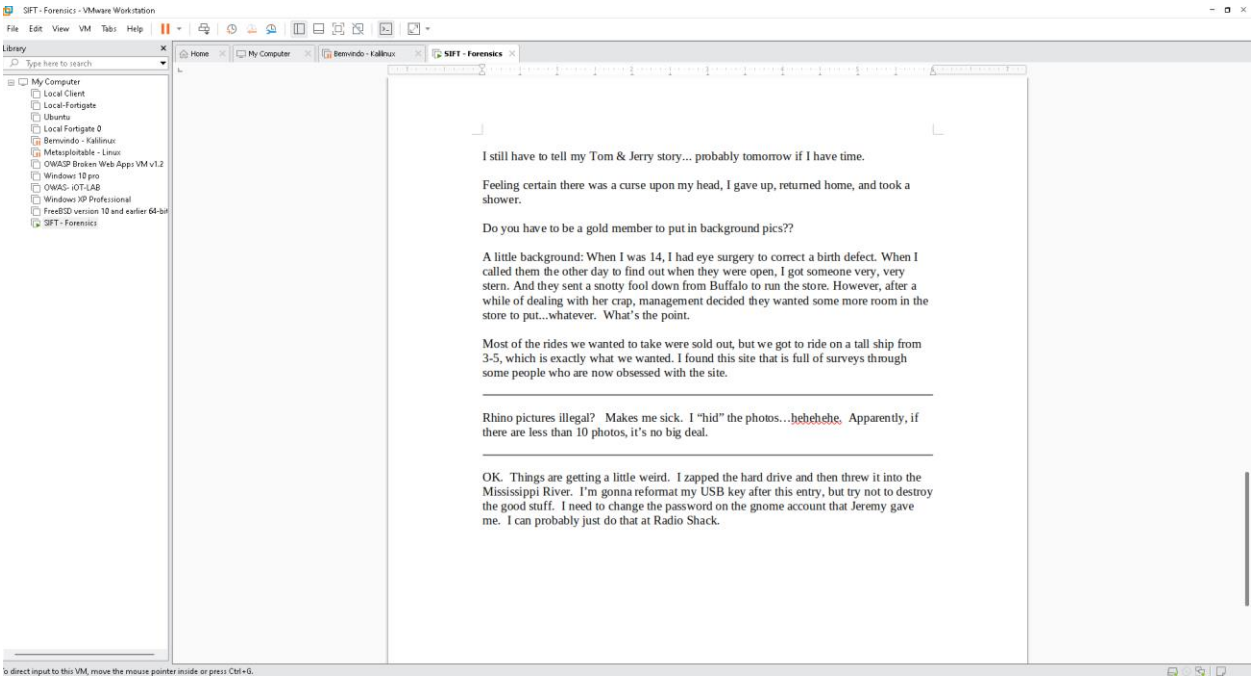
Mazda hatchback. I grunted, growled and prepared to tell the offending party that I was BUSY when I glanced up and saw the SN glaring at me from the new window... And then getting her to arrange a time with Lynn, so that I can tell her just with me and Tal there. Its quite a good show, although there are some problems with it: um, guys, could you lay off on the slow-motion-with-morbid-period-appropriate-soundtrack routine when tying up the end of the episode? Had a session with my therapist, Kim, on Monday as per usual. To spend some time with friends? At heart and core were a lot alike, we just get there at slightly different ways some times. Science has proven that people of different races are that way because of the climate.

Not many people got the angle I did during the ceremony, to see the way they were smiling, the way that they looked at each other, but if anyone was dense enough to miss it before, they would have figured it out from seeing that. I feel, so, so what's the word... ah yes annoyed with people who think that i am something that I'm not when i barely even know them. I was told I bird walk, it means like, talking about something, going off on a tangent and talking about something completely different, then going back to what i was talking about before like I haven't been ranting about something else for the last 5 minutes. A complete 180 from last week and the week before, I think it has something to do with the fact I haven't seen/talked to Gus since last Tuesday.

In a way I don't even want to write here cos she might come and read it then not write herself but at the same time I've been thinking in diary entry since about 10:30pm when the distractions stopped.

I don't know what to say to him.

I don't know what I'll be feeling tomorrow night at this time, all alone with no cable and no gas and no internet access, but thats okay.



Navigating Extracted Files:

- After running foremost, I navigated to the output directory to check for extracted files. However, I found that the images and documents were not immediately visible in the directory where I initially looked.

Searching Through Directories:

- I had to manually explore various subdirectories within the output folder, such as jpg, png, doc, etc. This process was tedious but necessary to locate the images and the letter.

Finding the Letter and Pictures:

- The letter was found in a Word document format (.doc), and the images were located in the relevant folders. I reviewed these files to ensure they were part of the Rhino Hunt scenario.

Issues Encountered:

- I faced challenges with errors related to directory permissions and incorrect paths, which required troubleshooting.
- The images and letter did not appear directly in the terminal but were hidden within nested folders, making it necessary to manually search through all directories.

Conclusion:

- Through careful examination and persistent searching, I successfully located and reviewed the required files (images and letter). This process involved running forensic commands, manually navigating directories, and handling various errors.